AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM10 (pm10)
Observed / expected baseline	272 / 34.0366 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-23 07:00 UTC
Location	29.7339, -95.2575
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 593	49.19 to 100; mean 77.5791	94.29 at 2026-06-23 06:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 593	25 to 34.4444; mean 29.3457	27 at 2026-06-23 06:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 593	0 to 230; mean 158.82	150 at 2026-06-23 06:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 594	0 to 10.8033; mean 3.8687	2.57222 at 2026-06-23 06:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5893.38 to 5954.67; mean 5934.03	5953.83 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	41.7736 to 1968.25; mean 854.297	552.684 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	35.0234 to 54.5962; mean 39.6063	38.334 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1004.61 to 1019.57; mean 1013.63	1015.61 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	17.0297 to 21.8722; mean 19.806	21.081 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-1.92568 to 2.43432; mean -0.1622	-0.874463 at 2026-06-23 06:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	1.15017 to 6.18109; mean 3.7798	3.16729 at 2026-06-23 06:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	17 / 1098	0 to 0.045; mean 0.0161	0.006 at 2026-06-23 07:00Z; at event time
openaq	PM10 (pm10)	ug/m3	3 / 217	18 to 272; mean 54.0046	272 at 2026-06-23 07:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM2.5 (pm25)	ug/m3	81 / 3263	0 to 42.1; mean 7.356	42.1 at 2026-06-23 07:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 17.4833	0 at 2026-06-23 07:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	57 to 95; mean 79.75	89 at 2026-06-23 07:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 1.27; mean 0.0172	0 at 2026-06-23 07:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1009 to 1020; mean 1015.66	1017 at 2026-06-23 07:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	25.46 to 34.74; mean 29.8001	27.3 at 2026-06-23 07:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 304; mean 169.872	175 at 2026-06-23 07:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 6.56; mean 2.3562	2.79 at 2026-06-23 07:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	67 / 4793	0 to 65.9; mean 8.7692	5.8 at 2026-06-23 07:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	4 / 4	0.0242468 to 0.0249866; mean 0.0246	0.0247994 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	6 / 6	9.43882e-05 to 0.000118622; mean 0.0001	0.000101698 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	1.683e-05 to 2.85183e-05; mean 0	2.64605e-05 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	6 / 6	-0.00020099 to 4.24095e-05; mean -0	1.36867e-05 at 2026-06-22 20:16Z; 10 h 44 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-22 20:16Z; 10 h 44 min before event
tceq	Carbon monoxide (co)	ppm	3 / 215	0.1 to 0.8; mean 0.2456	0.2 at 2026-06-23 07:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 838	-1.7 to 18.5; mean 3.1986	9.6 at 2026-06-23 07:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 214	-1.6 to 0.5; mean -0.4561	-0.2 at 2026-06-23 07:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-21 18Z 71.06, 06-22 00Z 85.81, 06Z 91.7, 12Z 73.85, 18Z 61.24, 06-23 00Z 83.29, 06Z 91.96, 12Z 69.79, 18Z 60.09, 06-24 00Z 81.04, 06Z 93.06, 12Z 70.97, 18Z 53.51
asos	Air temperature (temperature)	06-21 18Z 30.9, 06-22 00Z 27.96, 06Z 27.3, 12Z 30.71, 18Z 32.6, 06-23 00Z 28.26, 06Z 26.5, 12Z 30.27, 18Z 32.76, 06-24 00Z 28.26, 06Z 26, 12Z 30.06, 18Z 32.88
asos	Wind direction (wind_direction)	06-21 18Z 165.9, 06-22 00Z 155.4, 06Z 164.8, 12Z 174.4, 18Z 171.2, 06-23 00Z 151.1, 06Z 115.7, 12Z 141.9, 18Z 185.4, 06-24 00Z 164.9, 06Z 122.8, 12Z 183, 18Z 185
asos	Wind speed (wind_speed)	06-21 18Z 5.472, 06-22 00Z 3.925, 06Z 3.23, 12Z 4.925, 18Z 5.659, 06-23 00Z 3.929, 06Z 2.293, 12Z 3.976, 18Z 4.694, 06-24 00Z 3.109, 06Z 1.845, 12Z 3.29, 18Z 3.344
noaa_gfs	500-hPa geopotential height (gh_500)	06-22 00Z 5894, 06Z 5915, 12Z 5908, 18Z 5934, 06-23 00Z 5946, 06Z 5954, 12Z 5946, 18Z 5954, 06-24 00Z 5951, 06Z 5950, 12Z 5927, 18Z 5928
noaa_gfs	Planetary boundary layer height (pbl_height)	06-22 00Z 879, 06Z 552.2, 12Z 547, 18Z 1745, 06-23 00Z 1046, 06Z 452.6, 12Z 249.3, 18Z 1687, 06-24 00Z 961.3, 06Z 336.3, 12Z 251.8, 18Z 1544
noaa_gfs	Precipitable water (precipitable_water)	06-22 00Z 53.65, 06Z 43.63, 12Z 42.18, 18Z 39.6, 06-23 00Z 37.86, 06Z 38.28, 12Z 36.73, 18Z 38.35, 06-24 00Z 38.24, 06Z 35.83, 12Z 35.36, 18Z 35.56
noaa_gfs	Surface pressure (surface_pressure)	06-22 00Z 1007, 06Z 1009, 12Z 1010, 18Z 1013, 06-23 00Z 1012, 06Z 1015, 12Z 1017, 18Z 1017, 06-24 00Z 1015, 06Z 1017, 12Z 1016, 18Z 1015
noaa_gfs	850-hPa temperature (t_850)	06-22 00Z 20.18, 06Z 19.92, 12Z 18.85, 18Z 17.85, 06-23 00Z 20.48, 06Z 21.18, 12Z 20.73, 18Z 17.45, 06-24 00Z 20.45, 06Z 21.28, 12Z 20.26, 18Z 19.02
noaa_gfs	10-m eastward wind (u_10m)	06-22 00Z -0.9744, 06Z -0.617, 12Z -0.4874, 18Z -0.0208, 06-23 00Z -0.8663, 06Z -0.8745, 12Z -0.4436, 18Z -0.6389, 06-24 00Z 0.5653, 06Z 0.3389, 12Z 0.7901, 18Z 1.282
noaa_gfs	10-m northward wind (v_10m)	06-22 00Z 5.284, 06Z 4.083, 12Z 4.048, 18Z 5.175, 06-23 00Z 5.574, 06Z 3.237, 12Z 2.334, 18Z 4.014, 06-24 00Z 4.197, 06Z 2.914, 12Z 1.777, 18Z 2.72
openaq	Ozone (ozone)	06-21 18Z 0.02069, 06-22 00Z 0.01395, 06Z 0.01341, 12Z 0.01596, 18Z 0.01922, 06-23 00Z 0.01596, 06Z 0.01065, 12Z 0.0136, 18Z 0.02349, 06-24 00Z 0.0142, 06Z 0.01125, 12Z 0.01527, 18Z 0.02871
openaq	PM10 (pm10)	06-21 18Z 23.47, 06-22 00Z 34.44, 06Z 36.06, 12Z 55, 18Z 49.61, 06-23 00Z 45.06, 06Z 69.33, 12Z 70.56, 18Z 69.67, 06-24 00Z 48, 06Z 56.22, 12Z 76.83, 18Z 78.17
openaq	PM2.5 (pm25)	06-21 18Z 4.382, 06-22 00Z 5.88, 06Z 10.04, 12Z 13.92, 18Z 4.518, 06-23 00Z 4.748, 06Z 6.978, 12Z 7.383, 18Z 6.545, 06-24 00Z 6.306, 06Z 8.081, 12Z 8.709, 18Z 8.291
openweather	Cloud cover (cloud_cover)	06-21 18Z 77.92, 06-22 00Z 10.37, 06Z 7.2, 12Z 17.83, 18Z 5.31, 06-23 00Z 1.733, 06Z 2.419, 12Z 5.167, 18Z 4.793, 06-24 00Z 0.3333, 06Z 44.61, 12Z 37.17, 18Z 40.2
openweather	Relative humidity (humidity)	06-21 18Z 74.4, 06-22 00Z 84.23, 06Z 91.17, 12Z 80.9, 18Z 66.86, 06-23 00Z 79.73, 06Z 90.77, 12Z 78.87, 18Z 64.31, 06-24 00Z 76.33, 06Z 90.26, 12Z 79.43, 18Z 62.8
openweather	Precipitation rate (precipitation)	06-21 18Z 0.2472, 06-22 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-23 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-24 00Z 0, 06Z 0, 12Z 0, 18Z 0
openweather	Surface pressure (pressure)	06-21 18Z 1010, 06-22 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1014, 06-23 00Z 1016, 06Z 1017, 12Z 1020, 18Z 1018, 06-24 00Z 1018, 06Z 1018, 12Z 1018, 18Z 1018
openweather	Air temperature (temperature)	06-21 18Z 31.64, 06-22 00Z 28.69, 06Z 27.46, 12Z 30.33, 18Z 33.66, 06-23 00Z 29.38, 06Z 26.91, 12Z 29.94, 18Z 33.92, 06-24 00Z 29.79, 06Z 26.52, 12Z 29.5, 18Z 33.57
openweather	Wind direction (wind_direction)	06-21 18Z 175.5, 06-22 00Z 157.9, 06Z 163.4, 12Z 177.4, 18Z 174.3, 06-23 00Z 155.9, 06Z 145.3, 12Z 156, 18Z 199.6, 06-24 00Z 173.9, 06Z 159.4, 12Z 199.2, 18Z 193
openweather	Wind speed (wind_speed)	06-21 18Z 3.026, 06-22 00Z 2.593, 06Z 2.518, 12Z 3.077, 18Z 2.897, 06-23 00Z 2.534, 06Z 1.715, 12Z 1.853, 18Z 2.292, 06-24 00Z 2.358, 06Z 1.629, 12Z 2.051, 18Z 1.788
purpleair	PM2.5 (pm25)	06-21 18Z 6.141, 06-22 00Z 7.425, 06Z 13.79, 12Z 13.15, 18Z 5.284, 06-23 00Z 7.118, 06Z 9.358, 12Z 7.903, 18Z 6.266, 06-24 00Z 7.95, 06Z 10.26, 12Z 10.07, 18Z 8.75
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1

Source	Metric	Values
sentinel5p	CO column (s5p_co_column)	06-22 18Z 0.02489, 06-23 18Z 0.0243
sentinel5p	CO granules available (s5p_co_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-21 18Z 0.000112, 06-22 18Z 9.933e-05, 06-23 18Z 9.576e-05
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-21 18Z 1.683e-05, 06-22 18Z 2.646e-05, 06-23 18Z 2.852e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-21 18Z -0.0001202, 06-22 18Z 5.768e-06, 06-23 18Z 3.65e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-21 18Z 1, 06-22 18Z 1, 06-23 18Z 1
tceq	Carbon monoxide (co)	06-21 18Z 0.2, 06-22 00Z 0.2222, 06Z 0.2278, 12Z 0.2778, 18Z 0.2333, 06-23 00Z 0.2278, 06Z 0.2389, 12Z 0.3, 18Z 0.2647, 06-24 00Z 0.2389, 06Z 0.2389, 12Z 0.2667, 18Z 0.2333
tceq	Nitrogen dioxide (no2)	06-21 18Z 1.3, 06-22 00Z 2.099, 06Z 2.53, 12Z 3.778, 18Z 2.265, 06-23 00Z 2.596, 06Z 4.177, 12Z 4.777, 18Z 3.124, 06-24 00Z 2.525, 06Z 4.26, 12Z 4.811, 18Z 3.809
tceq	Sulfur dioxide (so2)	06-21 18Z -0.4533, 06-22 00Z -0.4, 06Z -0.4556, 12Z -0.2857, 18Z -0.4722, 06-23 00Z -0.5056, 06Z -0.5222, 12Z -0.5, 18Z -0.5176, 06-24 00Z -0.4333, 06Z -0.5222, 12Z -0.3944, 18Z -0.3833

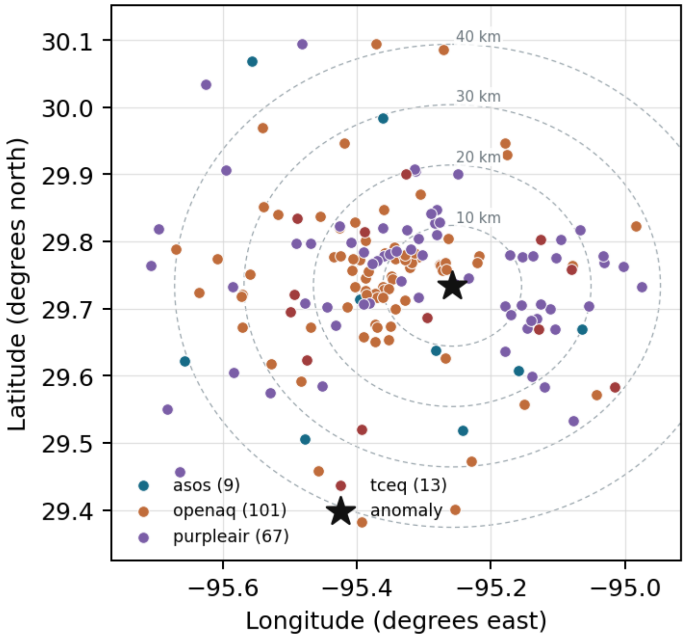
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

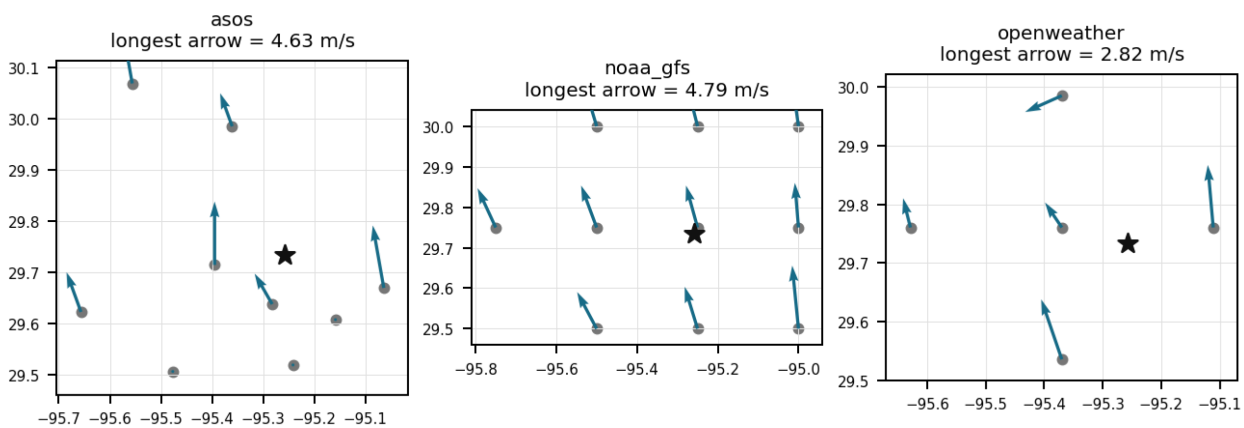
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (10.986 km) 77.2, MCJ (13.461 km) 78.52, EFD (17.008 km) 79.99, T41 (20.017 km) 77.93, LVJ (23.956 km) 77.23, IAH (29.579 km) 74.81, AXH (33.036 km) 75.65, SGR (40.495 km) 80.7, +1 more
asos	Air temperature (temperature)	HOU (10.986 km) 29.48, MCJ (13.461 km) 29.12, EFD (17.008 km) 29.18, T41 (20.017 km) 29.4, LVJ (23.956 km) 29.21, IAH (29.579 km) 29.88, AXH (33.036 km) 29.08, SGR (40.495 km) 29.08, +1 more
asos	Wind direction (wind_direction)	HOU (10.986 km) 169.6, MCJ (13.461 km) 162.5, EFD (17.008 km) 136.1, T41 (20.017 km) 178.2, LVJ (23.956 km) 151.9, IAH (29.579 km) 171.1, AXH (33.036 km) 121.5, SGR (40.495 km) 170.6, +1 more
asos	Wind speed (wind_speed)	HOU (10.986 km) 4.273, MCJ (13.461 km) 5.387, EFD (17.008 km) 4.191, T41 (20.017 km) 4.501, LVJ (23.956 km) 3.094, IAH (29.579 km) 3.615, AXH (33.036 km) 1.951, SGR (40.495 km) 4.366, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (1.931 km) 5934, gfs:29.75,-95.50 (23.481 km) 5934, gfs:29.75,-95.00 (24.925 km) 5934, gfs:29.50,-95.25 (26.019 km) 5934, gfs:30.00,-95.25 (29.598 km) 5934, gfs:29.50,-95.50 (35.014 km) 5934, gfs:29.50,-95.00 (36.001 km) 5934, gfs:30.00,-95.50 (37.713 km) 5934, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (1.931 km) 942.7, gfs:29.75,-95.50 (23.481 km) 985.2, gfs:29.75,-95.00 (24.925 km) 825.5, gfs:29.50,-95.25 (26.019 km) 698.6, gfs:30.00,-95.25 (29.598 km) 1054, gfs:29.50,-95.50 (35.014 km) 699, gfs:29.50,-95.00 (36.001 km) 790.5, gfs:30.00,-95.50 (37.713 km) 1099, +2 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (1.931 km) 39.62, gfs:29.75,-95.50 (23.481 km) 39.37, gfs:29.75,-95.00 (24.925 km) 40.01, gfs:29.50,-95.25 (26.019 km) 39.44, gfs:30.00,-95.25 (29.598 km) 39.93, gfs:29.50,-95.50 (35.014 km) 39.15, gfs:29.50,-95.00 (36.001 km) 39.81, gfs:30.00,-95.50 (37.713 km) 39.47, +2 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (1.931 km) 1014, gfs:29.75,-95.50 (23.481 km) 1013, gfs:29.75,-95.00 (24.925 km) 1015, gfs:29.50,-95.25 (26.019 km) 1015, gfs:30.00,-95.25 (29.598 km) 1013, gfs:29.50,-95.50 (35.014 km) 1014, gfs:29.50,-95.00 (36.001 km) 1016, gfs:30.00,-95.50 (37.713 km) 1011, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (1.931 km) 19.71, gfs:29.75,-95.50 (23.481 km) 19.79, gfs:29.75,-95.00 (24.925 km) 19.79, gfs:29.50,-95.25 (26.019 km) 20.05, gfs:30.00,-95.25 (29.598 km) 19.45, gfs:29.50,-95.50 (35.014 km) 20.08, gfs:29.50,-95.00 (36.001 km) 20.22, gfs:30.00,-95.50 (37.713 km) 19.53, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (1.931 km) -0.09636, gfs:29.75,-95.50 (23.481 km) -0.183, gfs:29.75,-95.00 (24.925 km) -0.1297, gfs:29.50,-95.25 (26.019 km) -0.1772, gfs:30.00,-95.25 (29.598 km) -0.1414, gfs:29.50,-95.50 (35.014 km) -0.4697, gfs:29.50,-95.00 (36.001 km) 0.01114, gfs:30.00,-95.50 (37.713 km) 0.2653, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (1.931 km) 3.596, gfs:29.75,-95.50 (23.481 km) 3.758, gfs:29.75,-95.00 (24.925 km) 3.609, gfs:29.50,-95.25 (26.019 km) 3.739, gfs:30.00,-95.25 (29.598 km) 3.605, gfs:29.50,-95.50 (35.014 km) 3.774, gfs:29.50,-95.00 (36.001 km) 4.477, gfs:30.00,-95.50 (37.713 km) 3.594, +2 more
openaq	Ozone (ozone)	3378 (0.0 km) 0.01284, 2039 (5.211 km) 0.01606, 325 (6.39 km) 0.01565, 2046 (10.787 km) 0.01237, 320 (12.079 km) 0.01184, 1319333 (14.037 km) 0.01872, 332 (14.341 km) 0.01907, 3214 (14.841 km) 0.01786, +9 more
openaq	PM10 (pm10)	13380082 (0.0 km) 66.57, 15852419 (7.015 km) 56.23, 271 (14.341 km) 39.18
openaq	PM2.5 (pm25)	3379 (0.0 km) 16.34, 15539682 (0.019 km) 9.398, 8967123 (0.061 km) 11.17, 8967479 (0.061 km) 9.92, 9489522 (2.875 km) 7.271, 8967065 (2.921 km) 8.007, 12178865 (3.7 km) 8.613, 9201803 (3.915 km) 8.593, +73 more
openweather	Cloud cover (cloud_cover)	grid:center (11.235 km) 15.56, grid:east (14.438 km) 17.56, grid:south (24.557 km) 15.08, grid:north (29.948 km) 20.81, grid:west (35.938 km) 18.42
openweather	Relative humidity (humidity)	grid:center (11.235 km) 79.26, grid:east (14.438 km) 82.6, grid:south (24.557 km) 79.56, grid:north (29.948 km) 78.11, grid:west (35.938 km) 79.22
openweather	Precipitation rate (precipitation)	grid:center (11.235 km) 0.01042, grid:east (14.438 km) 0.02611, grid:south (24.557 km) 0.02444, grid:north (29.948 km) 0.009861, grid:west (35.938 km) 0.015
openweather	Surface pressure (pressure)	grid:center (11.235 km) 1016, grid:east (14.438 km) 1016, grid:south (24.557 km) 1016, grid:north (29.948 km) 1016, grid:west (35.938 km) 1016
openweather	Air temperature (temperature)	grid:center (11.235 km) 30.1, grid:east (14.438 km) 29.58, grid:south (24.557 km) 29.58, grid:north (29.948 km) 30, grid:west (35.938 km) 29.74
openweather	Wind direction (wind_direction)	grid:center (11.235 km) 162.7, grid:east (14.438 km) 183.7, grid:south (24.557 km) 186.8, grid:north (29.948 km) 143.4, grid:west (35.938 km) 172.7
openweather	Wind speed (wind_speed)	grid:center (11.235 km) 2.348, grid:east (14.438 km) 3.687, grid:south (24.557 km) 1.926, grid:north (29.948 km) 2.078, grid:west (35.938 km) 1.742
purpleair	PM2.5 (pm25)	99797 (0.58 km) 7.348, 166435 (2.659 km) 12.14, 6752 (5.292 km) 7.819, 222601 (6.69 km) 10.12, 133994 (7.384 km) 0.6014, 235425 (8.236 km) 7.871, 229853 (8.533 km) 5.937, 127441 (8.715 km) 10.81, +59 more
tceq	Carbon monoxide (co)	482011035 (0.021 km) 0.1792, 482011039 (14.356 km) 0.1972, 482011052 (15.437 km) 0.3597
tceq	Nitrogen dioxide (no2)	482011035 (0.021 km) 6.046, 482010416 (6.399 km) 3.184, 482011039 (14.356 km) 2.343, 482010026 (14.86 km) 4.132, 482011052 (15.437 km) 8.276, 482011015 (17.424 km) 2.172, 482010024 (19.728 km) 1.152, 482011066 (22.748 km) 2.444, +4 more
tceq	Sulfur dioxide (so2)	482011035 (0.021 km) -0.1986, 482011039 (14.356 km) 0.1986, 482010051 (24.254 km) -1.394

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

The anomaly occurred on June 23rd, with the highest PM2.5 levels recorded at 07:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Local resuspension of road dust, construction dust, or other coarse mechanical particulate matter is a leading hypothesis because PM10 reached 272 ug/m3 while PM2.5 was much lower at 42.1 ug/m3 nearby, indicating a coarse-particle-dominant event rather than a purely fine-particle episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

Meteorological conditions in Houston favored accumulation of locally emitted dust because winds were light, humidity was near 90 to 94 percent, skies were mostly clear, surface pressure was high, and nighttime boundary-layer height was shallow near the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The Houston anomaly is most consistent with a local coarse-particle source because openaq recorded PM10 at 272 ug/m3 at 07:00 UTC while nearby PM2.5 responses were much smaller and less spatially coherent.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Meteorological trapping is supported because near the anomaly time Houston had very high surface humidity around 89 to 94 percent, light winds around 1.7 to 2.8 m/s, and a relatively shallow nighttime boundary layer near 553 m, which would favor accumulation of locally emitted or resuspended particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The large disparity between the severe local PM10 spike and modest regional fine PM2.5 readings indicates a localized mechanical dust or industrial emission event close to the openaq monitoring station.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

Local dust or mechanically generated coarse particles are supported because the PM10 anomaly reached 272 ug/m3 at the anomaly site while PM2.5 was elevated to 42.1 ug/m3 at the same time, indicating a particle mixture with a strong coarse fraction rather than a purely fine-particle event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

The anomaly is a result of a wildfire or agricultural burning in the surrounding areas, releasing large amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The nearest openaq ozone measurement at the same location and time was 0.006 ppm at 2026-06-23T07:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

NO_2 levels were also elevated, with a peak value of 9.6 ppb at monitor 482011035, located approximately 0.021 km from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

The PM2.5 levels exceeded the normal range, with a peak value of 67 ug/m3 at sensor 99797, located approximately 0.58 km from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The PM2.5 levels are elevated, with a mean of 8.7692 ug/m3 and a maximum value of 65.9 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The sharp dominance of coarse PM10 over fine PM2.5 under stagnant surface conditions indicates a localized fugitive dust or mechanical emission source near the monitoring site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The sharp contrast at 07:00 UTC on June 23, 2026, between the elevated PM10 concentration of 272.0 ug/m3 from OpenAQ and much lower PM2.5 values from PurpleAir and OpenAQ indicates that the anomaly was dominated by localized coarse-mode particulate matter.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The air quality anomaly in Houston, Texas on June 23rd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

An atmospheric PM10 anomaly occurred at coordinates (29.734, -95.257) on June 23, 2026, at 07:00 UTC, reaching a concentration of 272.0 ug/m3 compared to an expected baseline of 34.04 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Persistent southeasterly surface winds blowing around an upper-level high-pressure ridge transported regional coarse background aerosols into the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

Meteorological stagnation and shallow overnight mixing likely amplified near-surface particle accumulation in Houston because winds were weak at about 2 to 3 m/s, humidity was around 90 to 94 percent, surface pressure was high near 1016 to 1017 hPa, and modeled planetary boundary layer height was only about 450 to 550 m near the event time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The air quality anomaly in Houston, Texas is not caused by changes in temperature.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

Overnight atmospheric stability with low boundary layer height and low wind speeds suppressed vertical mixing, trapping coarse particulate emissions near the ground surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

Fresh regional smoke or sulfur-rich combustion is not the leading explanation because ozone was low at the anomaly time and co and so2 were not strongly elevated near the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Coinciding with the air quality anomaly at 07:00 UTC on June 23, 2026, surface wind speeds were low at approximately 2.57 m/s and modeled planetary boundary layer heights were restricted to between 249 m and 452 m.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

Moderate PM10 readings at surrounding OpenAQ stations and consistently low regional PM2.5 values across PurpleAir sensors indicate that the extreme PM10 spike was a localized event rather than a widespread regional dust plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Model atmospheric boundary layer heights from GFS dropped below 450 meters and surface wind speeds recorded by ASOS fell to approximately 2.3 m/s during the nocturnal hours of June 23, 2026, supporting nocturnal atmospheric trapping near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The air quality anomaly in Houston, Texas is not caused by wind direction or speed.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

A fresh combustion plume is a less likely primary explanation for the PM10 anomaly because ozone was suppressed while co, no2, and so2 were not strongly elevated enough to clearly indicate a major smoke or industrial combustion event at the same location and time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

Regional combustion smoke or sulfur-rich industrial plume dominance is contradicted because ozone was low at 0.006 ppm, carbon monoxide was only 0.2 ppm, sulfur dioxide was not elevated at the nearest monitor, and satellite carbon monoxide and sulfur dioxide columns were not strongly enhanced in the closest overpass window.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

The nearest openaq PM2.5 measurement at the same location and time was 42.1 ug/m3 at 2026-06-23T07:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

NOAA GFS model output and surface ASOS observations demonstrate a shallow nocturnal planetary boundary layer below 500 meters and light surface wind speeds under 2.6 m/s near the monitor at the time of the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The openaq PM10 measurement at 29.734, -95.257 was 272.0 ug/m3 at 2026-06-23T07:00:00+00:00, compared with an expected value of 34.03658536585366 ug/m3, corresponding to a z-score of 10.419528105824376 and classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

At 07:00 UTC on June 23, 2026, an OpenAQ sensor recorded a PM10 concentration spike of 272.0 ug/m3 in Houston, while surrounding PM2.5 sensors on OpenAQ and PurpleAir recorded low fine particulate concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

At the exact time and location of the PM10 anomaly, ambient PM2.5 concentrations peaked at 42.1 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.